SAI YASASWINI MAJETY

Senior Backend & Cloud Infrastructure Engineer | Distributed Systems Architect

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Professional Summary

- **6+ Years of Production Engineering Leadership:** Senior Software Engineer specializing in high-concurrency Python backend microservices, real-time telemetry streaming pipelines, cloud infrastructure automation, and zero-downtime database architectures.
- **Mission-Critical Aviation Telemetry (Viasat Inc.):** 5.7+ continuous years delivering Five-Nines platform availability (**99.999% SLA**) across **1,000+ commercial aircraft** (Delta Air Lines), advancing across 4 engineering cadres via rapid merit promotions.
- **High-Throughput Asynchronous Python Microservices:** Specialized in asynchronous Python (FastAPI, asyncio, uvloop, Pydantic v2, Django, Flask, Celery) developing high-volume RESTful APIs and distributed worker pipelines handling 12,000+ requests/second.
- **Zero-Downtime Database Architecture:** Proven production track record executing live online schema migrations on multi-terabyte transactional MySQL and PostgreSQL databases via trigger-based CDC and shadow tables, executing table cutovers in under **50 milliseconds**.
- **Enterprise PKI & Security Automation:** Engineered automated enterprise TLS/SSL certificate lifecycle daemons across 150+ edge endpoints and 1,000+ aircraft terminals, eliminating manual renewal outages and earning the **Viasat Spotlight Recognition Award**.
- **Edge Buffering & In-Flight Bandwidth Reduction:** Designed store-and-forward edge queueing daemons that prevented telemetry loss during satellite spot-beam handovers while cutting airborne uplink bandwidth consumption by **65%** via Zstandard compression.
- **Identity Federation & Partner Systems Integration:** Architected decoupled OAuth 2.0 and SAML/Okta identity gateways integrating partner airline systems, reducing third-party partner onboarding lead time from **3 weeks to 2 days**.
- **Published Systems Research & IP Innovation:** Co-inventor of published **Indian Patent 202541026299** (*Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning*), cutting continual neural network fine-tuning compute requirements by **80%**.
- **Executive Recognition & Corporate Honors:** Awarded **2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg, 11 formal corporate citations, and 1st Place out of 24 international teams in the Viasat IoT Hackathon.

Flagship Honors, Awards & Technical Credentials

- **Co-Inventor, Indian Patent Published (App. No. 202541026299):** *Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning*, cutting neural network retraining compute by **80%** via parameter-efficient adapter routing.
- **Viasat CEO Performance Awards (2x Recipient):** Commended directly by Chairman & CEO Mark Dankberg for outstanding operational reliability and telemetry execution across the Delta Air Lines commercial passenger fleet.
- **Viasat Spotlight Recognition Award:** Conferred for enterprise technical innovation in automating fleet-wide PKI certificate rotation and zero-touch cryptographic key deployment across commercial aviation fleets.
- **Viasat Architecture Commendation & Engineering Excellence:** Commended by senior engineering leadership for decoupled OAuth 2.0 / SAML identity gateways and zero-downtime Multi-AZ database cluster migrations.
- **1st Place Winner, Viasat Global IoT Hackathon:** Ranked 1st out of 24 international engineering teams for conceiving and deploying a real-time smart edge sensor telemetry platform with live web dashboarding within 24 hours.
- **Dean's Merit Scholarship Distinction:** Conferred by SASTRA Deemed University for ranking in the top 3% academic cohort across the B.Tech program.

Technical Skills Matrix

Programming Languages: Python 3.11/3.12, Go, JavaScript (ES6+), TypeScript, SQL, Bash/Shell Scripting, C/C++, Groovy
Backend Frameworks & APIs: FastAPI, Pydantic v2, Django, Flask, Celery, SQLAlchemy, Alembic, asyncio, uvloop, RESTful APIs, gRPC, mTLS
Cloud Infrastructure & DevOps: AWS (ECS, RDS Multi-AZ, S3, IAM, CloudWatch, KMS, CloudFormation, Route53), Docker, Kubernetes, Jenkins, Terraform, Helm
Databases & Distributed Caching: PostgreSQL, PostGIS, MySQL 8.0, MariaDB, TimescaleDB, SQLite (WAL mode), Redis In-Memory Cache, ElastiCache
Streaming, Messaging & Telemetry: Apache Kafka, RabbitMQ, AWS SQS, AWS SNS, MQTT, Server-Sent Events (SSE), WebSockets, Store-and-Forward Buffers
Security, Identity & PKI: OAuth 2.0, Okta, SAML 2.0, Automated TLS/SSL X.509 Lifecycle, Public Key Infrastructure (PKI), HashiCorp Vault, Secrets Manager
Testing & Observability: PyTest, Robot Framework, Postman, Playwright, CloudWatch Metrics & Alarms, Splunk, Datadog, SonarQube, Synthetic Probes
AI Platforms & Data Engineering: Retrieval-Augmented Generation (RAG), FAISS Vector Indexing, LiteLLM Semantic Caching, Hugging Face, LangChain

Professional Experience

Independent Applied Research & Systems Engineering — **Houston, TX**

AI Systems & Cloud Infrastructure Researcher — *Mar 2024 - Present*

Environment: Python 3.12, FastAPI, Pydantic v2, PostgreSQL, PostGIS, Redis, Docker, AWS (EC2, S3, RDS), Git, PyTest, Linux, Terraform

- **Co-invented Indian Patent 202541026299** (*Modular Deep Learning Architecture for Cross-Domain Transfer and Incremental Learning*), designing dynamically routable adapter modules attached to frozen base networks that cut fine-tuning compute requirements by **80%**.
- **Architected sub-20ms FastAPI semantic caching microservices** utilizing asynchronous connection pooling, Redis in-memory vector storage, and Pydantic v2 serialization, sustaining **12,000+ requests/second** under peak synthetic loads.
- **Engineered store-and-forward edge telemetry daemon (edge-telemetry-daemon)** in Python 3.11 with SQLite WAL ring buffering and Zstandard streaming compression, guaranteeing zero telemetry loss during network dropouts while cutting uplink payload size by **65%**.
- **Architected real-time geospatial infrastructure monitoring platform (geospatial-infrastructure-monitoring-saas)** in FastAPI and PostGIS, rendering sub-500ms Mapbox vector tiles for 100,000+ assets and automating EAM maintenance dispatch (IBM Maximo/SAP PM bridge).
- **Engineered automated database benchmarking harnesses** in PostgreSQL and PostGIS to measure query latency deltas across spatial GiST indexes, reducing execution times on complex spatial containment joins by **45%**.
- **Authored modular Terraform infrastructure-as-code blueprints** and Docker Compose environments automating multi-container microservice deployments with strict IAM least-privilege boundary policies and zero-trust validation.

Viasat Inc. : Commercial Aviation Platform & Security Infrastructure — **Chennai, India / Carlsbad, CA**

Senior Software Engineer - I (Client: Delta Air Lines Commercial Fleet) — *Jan 2023 - Feb 2024*

Environment: Python 3.9/3.10, FastAPI, Django, PostgreSQL, MySQL 8.0 Multi-AZ, Apache Kafka, Redis, Docker, Kubernetes, AWS, Linux, OpenSSL, PKI

- **Sustained a five-nines (99.999%) availability SLA** across 1,000+ commercial passenger aircraft for Delta Air Lines, engineering distributed backend services in Python that processed continuous avionics, RF link, and network telemetry streams.
- **Executed zero-downtime online schema refactoring** on multi-million row operational MySQL and PostgreSQL databases via trigger-based CDC and shadow tables, completing global schema switchovers with under **50 milliseconds** of table locking.
- **Automated enterprise PKI certificate lifecycle daemon** across 1,000+ aircraft terminals, orchestrating automated X.509 renewal handshakes and cryptographic rotation that earned the prestigious **Viasat Spotlight Recognition Award**.
- **Engineered edge store-and-forward queueing daemons** in Python that buffered telemetry locally on airborne servers during satellite spot-beam handovers, eliminating telemetry loss and reducing uplink bandwidth consumption by **65%**.
- **Decoupled high-velocity telemetry ingestion** from analytical persistence layers by implementing Apache Kafka message buffers, eliminating database write bottlenecks during peak transcontinental flight banks.
- **Architected secure OAuth 2.0 and SAML federation gateways** integrating airline identity providers (Delta SkyMiles) with in-cabin network authentication services, supporting over **250,000 daily passenger authentications** and cutting partner onboarding from **3 weeks to 2 days**.
- **Engineered customer-facing fleet health monitoring portals** using Python, React, and WebSocket telemetry streams, enabling flight operations centers to visualize real-time aircraft coordinates, link margins, and latency anomalies across global flight corridors.
- **Built high-availability database failover procedures** for AWS RDS Multi-AZ instances with read replicas, ensuring automated recovery and data consistency during unplanned regional network outages.
- **Awarded 2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg for exceptional operational stability during large-scale airline fleet software rollouts.

Viasat Inc. : Cloud Platform & API Microservices — **Chennai, India**

Software Engineer II (Multi-Region Cloud Deployment & Test Systems) — *Jan 2021 - Dec 2022*

Environment: Python 3.8/3.9, FastAPI, AWS (CloudFormation, RDS, ECS, S3, IAM), Jenkins, Docker, Linux, Shell Scripting, PyTest

- **Accelerated multi-region cloud deployment velocity by 65%** (from 4 hours to 45 minutes MTTR) by authoring modular AWS CloudFormation IaC templates and declarative Jenkins CI/CD pipelines across staging and production clusters.
- **Elevated test coverage from 60% to >92%** across critical avionics, billing, and session authorization microservices by architecting automated PyTest suites, synthetic health harnesses, and network failure simulations.
- **Engineered automated cross-region replication monitoring** and health failover probes for enterprise AWS RDS MySQL instances, guaranteeing transactional recovery objectives (RPO/RTO) under disaster recovery drills.
- **Awarded the 2021 CEO Mark Dankberg Performance Bonus** for high-impact infrastructure delivery and zero-downtime fleet reliability.

Professional Experience (Continued)

Viasat Inc. : Distributed Systems, Search & Tooling Automation — **Chennai, India**

Software Engineer I & Software Engineer Intern — *Jan 2018 - Dec 2020*

Environment: Python 3.7/3.8, Django, Flask, PostgreSQL, MariaDB, Sphinx, Robot Framework, Docker, Jenkins, Git, Linux

- **Commercial Air PAT (Pipeline Observability & Quality Dashboard):** Engineered real-time observability platform in Python Flask and React aggregating build results and defect telemetry across **50+ Jenkins nodes**, cutting release verification latency by **40%**.
- **CandEx 2.0 Platform (Internal Talent Operations & Search Engine):** Accelerated technical resume search query response times from **3.2s to <180ms** (17x speedup) by integrating Sphinx full-text search engine with Python Django microservices across 10,000+ records.
- **Automated pre-merge regression verification pipelines** in Groovy and Robot Framework, executing 100+ automated test scenarios per release candidate and preventing breaking API regressions from reaching staging environments.
- **Engineered automated ETL data extraction pipelines** that normalized heterogeneous sensor logs into PostgreSQL, accelerating daily operational reporting generation from **4 hours to 15 minutes**.
- **Built responsive web interfaces** using React and JavaScript to display real-time terminal connection status, signal-to-noise ratios (SNR), and historical uptime trends for network operations engineers.
- **Won 1st Place out of 24 international engineering teams in the Viasat Global IoT Hackathon** for designing an end-to-end connected IoT sensor prototype with cloud telemetry ingestion and live web dashboarding in 24 hours.
- **Recognized with formal Viasat Leadership Appreciation Award** and received direct conversion offer from intern to full-time SWE I.

Flagship Systems Architectures & Technical Projects

Real-Time Geospatial Critical Infrastructure Monitoring SaaS Platform — **FastAPI** | **PostGIS** | **Mapbox GL JS** | **Docker**

- Architected an end-to-end cloud platform in Python (FastAPI) and PostgreSQL/PostGIS to ingest high-frequency environmental sensor telemetry from 10,000+ spatial coordinates, generating dynamic Mapbox vector tiles (.pbf) with sub-500ms rendering for 100,000+ assets.
- Implemented an automated EAM integration bridge connecting sensor threshold alarms to enterprise work order systems (IBM Maximo / SAP PM), automatically generating preventative maintenance inspection tickets during threshold breaches.

Edge Store-and-Forward Telemetry Daemon — **Python 3.11** | **SQLite WAL** | **Zstandard** | **Asyncio** | **Systemd**

- Engineered an asynchronous local queueing engine for intermittent networks using SQLite WAL ring buffering, guaranteeing zero telemetry loss during link dropouts while cutting uplink bandwidth consumption by **65%** via Zstandard streaming compression.
- Architected crash-resilient queue persistence ensuring ACID durability and atomic batch drain recovery upon satellite uplink reconnection.

High-Throughput Semantic Vector Caching Microservice — **FastAPI** | **Redis** | **LiteLLM** | **Pydantic v2** | **Docker**

- Engineered a sub-20ms semantic caching proxy in FastAPI intercepting API queries with Redis vector indexing, achieving an 85% cache hit rate, reducing downstream database load by **60%**, and sustaining 12,000+ requests/second.
- Implemented two-pass semantic hashing with cosine distance similarity thresholds and TTL-based invalidation, eliminating redundant LLM API calls and enforcing strict token governance.

Open-Source Enterprise RAG Assistant (rag-hf-chatbot) — **Python 3.11** | **Hugging Face** | **FAISS** | **Streamlit**

- Engineered an interactive document question-answering system using Hugging Face embeddings and FAISS vector indexing, achieving sub-200ms semantic retrieval over complex documentation with structured source grounding.
- Implemented conversational context memory management and structured source citation grounding, eliminating model hallucination across multi-turn user dialogues.

Education & Academic Credentials

SASTRA Deemed University — **Thanjavur, India**

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering (EEE) — *Aug 2014 - May 2018*

Academic Distinction: Cumulative GPA: **8.2 / 10.0** | Recipient of **Dean's Merit Scholarship** (Top 3% cohort distinction).
Core Coursework: Microprocessor & Microcontroller Architectures, Control Systems, Data Structures, OOP (Python, C++), Relational Databases & SQL.
Campus Leadership & Capstone: Lead Technical Organizer, DAKSH National Techno-Management Fest; Capstone: PIC Microcontroller Telemetry System (ECIL Certified).